Prevalence and trends in tobacco use among adolescents aged 13–15 years in 143 countries, 1999–2018: findings from the Global Youth Tobacco Surveys

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Summary

Background Tobacco use is a leading preventable cause of morbidity and mortality worldwide. Little is known about recent prevalence and trends in tobacco use among adolescents globally. We aimed to assess the recent global prevalence of tobacco use in young adolescents and the secular trends in prevalence between 1999 and 2018.

Methods We used the most recent Global Youth Tobacco Surveys data on adolescents aged 13–15 years from 143 countries or territories that had done at least one survey between Jan 1, 2010, and Dec 31, 2018, to assess the recent prevalence of tobacco use; and data from 140 countries that had done two or more surveys between Jan 1, 1999, and Dec 31, 2018, to assess the trends in the prevalence of tobacco use.

Findings 530 234 adolescents were included from the 143 countries that had done at least one survey between 2010 and 2018. 1 192 312 adolescents were included from the 140 countries that had done two or more surveys between 1999 and 2018. The most recent global prevalence of cigarette smoking was 11·3% (95% CI 10·3–12·3) in boys and 6·1% (5·6–6·6) in girls, based on cigarette smoking on at least 1 day during the past 30 days, 6·0% (5·5–6·6) and 2·6% (2·4–2·9) based on smoking on at least 3 days, and 4·2% (3·8–4·6) and 1·6% (1·4–1·8) based on smoking on at least 6 days. The most recent prevalence of the use of tobacco products other than cigarettes (eg, chewing tobacco, snuff, dip, cigars, cigarillos, pipe, electronic cigarettes) on at least 1 day during the past 30 days was 11·2% (9·9–12·6) in boys and 7·0% (6·4–7·7) in girls. The most recent prevalence of any tobacco use on at least 1 day during the past 30 days was 17·9% (16·1–19·6) in boys and 11·5% (10·5–12·4) in girls. The prevalence of cigarette smoking on at least 1 day during the past 30 days decreased between the first and last surveys in 80 (57·1%) of 140 countries, was unchanged in 39 countries (27·9%), and increased in 21 countries (15·0%). However, the prevalence of the use of tobacco products other than cigarettes was unchanged or increased in 81 (59·1%) of 137 countries.

Interpretation The global prevalence of tobacco use among adolescents aged 13–15 years was substantial. Although the prevalence of cigarette smoking decreased over time in the majority of countries, the prevalence of the use of other tobacco products increased or did not change in the majority of countries during the past two decades. These findings re-emphasise the need to strengthen tobacco control efforts among young adolescents globally.

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See Comment page 234

For the Chinese translation of the abstract see Online for appendix 1

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For more on **WHO Framework Convention on Tobacco Control** see https://www.who.int/fctc/signatories_parties/en/

Introduction

Tobacco use is a major preventable risk factor for several cancers, cardiovascular diseases, and respiratory diseases. More than 8 million people worldwide die from tobacco use each year, and as many as 170·9 million disability-adjusted life-years lost were attributed to tobacco use in 2017.[1] The WHO Framework Convention on Tobacco Control (FCTC), which is a powerful legal instrument to address tobacco control both at the national level and globally, has been adopted by 182 countries and ratified by 168 as of October, 2020. The WHO Global Action Plan for the Prevention and Control of Noncommunicable Diseases 2013–20 has set the target to reduce the prevalence of tobacco use by 30% among individuals aged 15 years or older between 2010 and 2025. The 2030 Sustainable Development Agenda urges countries to fully implement tobacco control measures prescribed by the FCTC. Although the prevalence of tobacco use has globally decreased among adults in recent years,[2] tobacco use among both adults and children remains a major concern for public health.[3]

Tobacco use among adolescents and children is a problem of crucial importance, given that most adult smokers start smoking in their childhood or adolescent years[4] and tobacco is highly addictive.[5] In addition to the long-term health consequences of tobacco use, such as cancer, heart disease, and pulmonary diseases, which generally become apparent only after many years of smoking, other illnesses including respiratory diseases can affect young smokers.[4] Moreover, the younger people start smoking, the greater the mortality.[6]

A review of the Global Youth Tobacco Surveys (GYTS) in 131 countries or territories (hereafter referred to as countries) between 1999 and 2005 found that 8·9% of

For more on **WHO Global Action Plan for the Prevention and Control of Noncommunicable Diseases 2013–20** see https://apps.who.int/iris/bitstream/handle/10665/94384/9789241506236_eng.pdf;jsessionid=EC052A3996942E19BF1B3F598A590E46?sequence=1

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For more on **GYTS** see https://www.cdc.gov/tobacco/global/gtss/gtssdata/index.html

Research in context

Evidence before this study

We searched PubMed on Oct 10, 2020, to identify relevant papers on the global prevalence of cigarette smoking or tobacco use in children or adolescents published between Jan 1, 2000, and Oct 10, 2020, using the terms: ("tobacco use" OR "cigarette smoking") AND ("children" OR "adolescents" OR "youths"), with no language restrictions. Three previous reviews had assessed the prevalence globally: a review of the Global Youth Tobacco Surveys (GYTS) among adolescents aged 13–15 years from 131 countries or territories in 1999–2005; a review of the Global School-based Student Health Surveys (GSHS) among adolescents aged 12–15 years from 68 low-income and middle-income countries in 2006–13; and a review of GYTS among adolescents aged 13–15 years from 100 countries or territories to assess the trends in tobacco use from 1999 to 2008. However, none of these previous reviews estimated the prevalence on the basis of different frequencies of smoking (eg, cigarette use on ≥1 day, ≥3 days, or ≥6 days during the past 30 days) or included data after 2013.

Added value of this study

To our knowledge, our study provides the most recent analysis of global prevalence and trends of tobacco use in adolescents using the latest GYTS data (up to 2018). We found that the prevalence of cigarette smoking among adolescents aged 13–15 years was high globally, but it decreased between the first and last surveys in the majority of countries included in this study. However, the most recent global prevalence of use of tobacco products other than cigarettes was as high as that of cigarette smoking, and it was unchanged or increased in the majority of countries between their first and last surveys. Moreover, the prevalence of tobacco use decreased more in countries that had ratified the WHO Framework Convention on Tobacco Control (FCTC) than in countries that had not.

Implications of all the available evidence

Tobacco use among adolescents remains a serious public health issue worldwide. Further studies should examine which specific tobacco products other than cigarettes are used at the country level to guide adequate interventions. These findings suggest that tobacco control efforts among adolescents should be strengthened globally, and support a comprehensive implementation and adequate enforcement of the tobacco control measures described by the FCTC.

adolescents aged 13–15 years had smoked cigarettes on at least 1 day during the past 30 days and 11·2% had used a tobacco product other than cigarettes during the past 30 days.[7] A review of data from the Global School-based Student Health Surveys (GSHS) in 68 low-income and middle-income countries between 2006 and 2013 showed similar findings.[8] However, to our knowledge, no previous review has assessed recent secular trends in tobacco use or examined the prevalence of tobacco use on the basis of smoking cigarettes at different frequencies (ie, smoking cigarettes on ≥3 days or on ≥6 days during the past 30 days). More frequent tobacco use might better represent sustained tobacco usage, as opposed to sole cigarette experimentation.[9] Also, no previous review has assessed whether the secular trends in the prevalence of tobacco use are associated with country income (an indirect index of tobacco purchasing power for individuals) or with the country's FCTC ratification status (a proxy for comprehensiveness of a country's tobacco control policy).

We aimed to assess the global prevalence of cigarette smoking and the use of tobacco products other than cigarettes among young adolescents, the secular trends in tobacco use between 1999 and 2018, and whether the prevalence of tobacco use and the secular trends in tobacco use were associated with country income or FCTC ratification status.

Methods

Study design and participants

The GYTS is a global surveillance tool for tobacco use and related factors based on population-based samples of school-attending young adolescents aged 13–15 years. The same sampling strategy and standard self-administered and anonymous questionnaire are used in all participating countries. Detailed methods of the GYTS are described on the US Centers for Disease Control and Prevention (CDC) website. In brief, a two-stage sampling strategy was used in all countries. In the first stage, schools were randomly selected in the considered population (national or subnational), with sampling probability proportional to enrolment size, and classes within the selected schools were randomly chosen in the second stage. All selected students were eligible to participate in any GYTS questionnaire. The questionnaire, which uses the same core questions and possible closed-ended answers, was translated into the local languages for that country and back-translated to check accuracy. GYTS questionnaires are expected to be repeated at regular intervals in countries willing to do them. GYTS questionnaires in all countries were validated by the CDC and approved at the country level. The GYTS do not include any data that allow participant identification. The country datasets are publicly available and have complied with a corresponding national ethical board review.

We used the most recent data from 143 countries that had done at least one survey between 2010 and 2018, to assess the recent prevalence of tobacco use, and data from 140 countries that had done two or more surveys between 1999 and 2018, to assess the trends in the prevalence of tobacco use. The minority of adolescents younger than 13 years or older than 15 years in the GYTS dataset were excluded from our analysis.

	Number of countries	≥1 day					≥3 days					≥6 days				
		Boys		Girls		p value (boys vs girls)	Boys		Girls		p value (boys vs girls)	Boys		Girls		p value (boys vs girls)
		Prevalence	p value	Prevalence	p value		Prevalence	p value	Prevalence	p value		Prevalence	p value	Prevalence	p value	
Total	143	11·3% (10·3–12·3)	..	6·1% (5·6–6·6)	..	<0·0001	6·0% (5·5–6·6)	..	2·6% (2·4–2·9)	..	<0·0001	4·2% (3·8–4·6)	..	1·6% (1·4–1·8)	..	<0·0001
Age	..	..	<0·0001*	..	<0·0001*	..	..	<0·0001*	..	<0·0001*	..	..	<0·0001*	..	<0·0001*	..
13 years	143	6·8% (6·2–7·5)	..	3·4% (3·1–3·8)	..	<0·0001	3·0% (2·6–3·3)	..	1·5% (1·3–1·7)	..	<0·0001	1·9% (1·6–2·1)	..	0·9% (0·8–1·0)	..	<0·0001
14 years	143	10·1% (9·2–11·1)	..	5·7% (5·1–6·2)	..	<0·0001	5·0% (4·5–5·4)	..	2·3% (2·0–2·6)	..	<0·0001	3·3% (2·9–3·6)	..	1·4% (1·2–1·6)	..	<0·0001
15 years	143	15·4% (14·1–16·7)	..	8·7% (7·8–9·5)	..	<0·0001	9·0% (8·2–9·8)	..	3·7% (3·3–4·0)	..	<0·0001	6·5% (5·9–7·1)	..	2·6% (2·2–2·9)	..	<0·0001
WHO region																
Africa	22	8·6% (6·8–10·5)	Ref	3·7% (2·5–4·9)	Ref	<0·0001	4·1% (3·1–5·1)	Ref	1·0% (0·7–1·2)	Ref	<0·0001	2·3% (1·7–2·9)	Ref	0·4% (0·3–0·6)	Ref	<0·0001
Americas	30	8·1% (6·7–9·5)	0·34†	6·3% (5·2–7·4)	0·0009†	0·024	3·6% (2·9–4·4)	0·22†	2·4% (1·9–2·9)	<0·0001†	0·0045	2·1% (1·7–2·6)	0·30	1·2% (0·9–1·5)	<0·0001†	0·0006
Eastern Mediterranean	24	12·2% (9·5–14·8)	0·015†	3·4% (2·6–4·1)	0·34†	<0·0001	7·0% (5·5–8·5)	0·0008†	1·2% (0·8–1·5)	<0·0001†	<0·0001	4·7% (3·7–5·8)	<0·0001†	0·7% (0·5–0·9)	0·0093†	<0·0001
Europe	33	10·6% (8·6–12·6)	0·075†	9·0% (7·4–10·5)	<0·0001†	0·11	7·4% (5·8–9·0)	0·0003†	5·5% (4·6–6·5)	<0·0001†	0·023	5·9% (4·6–7·3)	<0·0001†	4·0% (3·3–4·7)	<0·0001†	0·0072
South-East Asia	9	15·7% (10·5–20·9)	0·0058†	4·4% (2·5–6·2)	0·27†	<0·0001	6·4% (4·0–8·8)	0·042†	1·4% (0·7–2·2)	0·16†	<0·0001	4·2% (2·4–6·0)	0·025†	0·8% (0·3–1·2)	0·049†	0·0002
Western Pacific	25	17·6% (14·2–21·1)	<0·0001	7·1% (6·1–8·1)	<0·0001†	<0·0001	9·1% (7·1–11·1)	<0·0001†	2·6% (2·0–3·3)	<0·0001†	<0·0001	6·1% (4·7–7·5)	<0·0001†	1·5% (1·1–2·0)	<0·0001†	<0·0001
World Bank income category																
Low	18	10·5% (7·5–13·5)	Ref	3·0% (2·1–3·9)	Ref	<0·0001	5·2% (3·7–6·6)	Ref	0·7% (0·4–1·0)	Ref	<0·0001	2·8% (2·0–3·7)	Ref	0·4% (0·2–0·5)	Ref	<0·0001
Middle	41	12·3% (10·3–14·3)	0·16‡	4·9% (3·9–5·8)	0·0022‡	<0·0001	5·7% (4·8–6·6)	0·28‡	1·6% (1·3–1·9)	<0·0001‡	<0·0001	3·5% (2·9–4·2)	0·10‡	0·9% (0·7–1·1)	<0·0001‡	<0·0001
Upper middle	49	10·1% (8·9–11·4)	0·40‡	6·0% (5·1–6·8)	<0·0001‡	<0·0001	5·5% (4·6–6·4)	0·37‡	2·7% (2·2–3·2)	<0·0001‡	<0·0001	3·7% (3·1–4·3)	0·045‡	1·7% (1·3–2·0)	<0·0001‡	<0·0001
High	35	12·1% (9·9–14·3)	0·20‡	9·3% (7·7–11·0)	<0·0001‡	0·023	7·6% (6·0–9·3)	0·016‡	5·1% (4·1–6·0)	<0·0001‡	0·0050	5·9% (4·6–7·3)	<0·0001‡	3·4% (2·7–4·0)	<0·0001‡	0·0005
FCTC ratification status	..	..	0·022	..	0·20	..	..	0·012	..	0·020	..	..	0·047	..	0·14	..
Not ratified	18	14·7% (11·1–18·2)	..	6·7% (5·1–8·2)	..	<0·0001	8·4% (6·2–10·6)	..	2·0% (1·4–2·6)	..	<0·0001	5·3% (4·0–6·7)	..	1·3% (0·8–1·8)	..	<0·0001
Ratified	125	10·9% (9·9–11·9)	..	6·0% (5·5–6·6)	..	<0·0001	5·8% (5·3–6·4)	..	2·7% (2·4–3·0)	..	<0·0001	4·1% (3·7–4·5)	..	1·6% (1·4–1·8)	..	<0·0001

Prevalence is shown as % (95% CI). *p value for trend across different ages. †p value compared with the WHO African region as the reference. ‡pvalue compared with low-income countries as the reference.FCTC=Framework Convention on Tobacco Control.

Table 1: **Prevalence of cigarette smoking among adolescents aged 13–15 years by sex and frequency of cigarette smoking (number of smoking days during the past 30 days)**

Procedures

Current cigarette smoking was defined as smoking a cigarette on at least 1 day during the past 30 days. In response to the survey question "During the past 30 days, on how many days did you smoke cigarettes?", participants could choose from the following frequency categories: 0 days; 1 or 2 days; 3–5 days; 6–9 days; 10–19 days; 20–29 days; or all 30 days. Current other tobacco use was defined as the use of any form of tobacco products other than cigarettes within the past 30 days. In response to the survey question "During the past 30 days, have you ever used any form of tobacco products other than cigarettes (eg, chewing tobacco, snuff, dip, cigars, cigarillos, pipe, e-cigarettes)?", participants could choose either yes or no (hence responses did not allow for assessment of the frequency of other tobacco product use). Current any tobacco use was defined as using either cigarettes or other tobacco products on at least 1 day during the past 30 days.

The year that each country had ratified the WHO FCTC was extracted from the WHO FCTC website. Country income category was based on the World Bank classification corresponding to the most recent survey year of the GYTS in our study.

Statistical analysis

Prevalence estimates and 95% CIs were calculated using sampling weights, strata, and primary sampling units according to the GYTS methodology used in each country. Data analyses were done using the Complex Samples module in SPSS version 16.0. Estimates in each of the subgroups by WHO region, country income level, and FCTC ratification status were calculated using meta-analyses (Stata version 11.0) based on a random-effects model because of between-country heterogeneity (I^2 > 50%). Differences in prevalence between sexes, WHO regions, country income levels, or FCTC ratification status were classified as significant when the p value was less than 0·05. We used the trend χ^2 test to test for differences in prevalence over calendar years, and we accounted for all different periods of the available data in each country when calculating p values for trends. Secular trends in the prevalence of tobacco use were calculated per 10 calendar years, due to the small changes as estimated with a 1-year unit. For trends, p values less than 0·05 indicate significant downward or upward trends in prevalence.

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Results

530 234 adolescents were included from the 143 countries that had done at least one survey between 2010 and 2018 (appendix 2 pp 1–4); 257 118 boys (48·5%) and 273 116 girls (51·5%). Of these 143 countries, 22 (15·4%) were located in the WHO African region; 30 (21·0%) in the Americas; 24 (16·8%) in Eastern Mediterranean; 33 (23·1%) in Europe; nine (6·3%) in South-East Asia; and 25 (17·5%) in Western Pacific.

See Online for appendix 2

1 192 312 adolescents were included from the 140 countries that had done two or more surveys between 1999 and 2018 (appendix 2 pp 5–17); 565 588 boys (47·4%) and 626 724 girls (52·6%). Within this period, 53 (37·9%) of 140 countries had done two surveys, 45 (32·1%) had done three surveys, 35 (25·0%) had done four surveys, five (3·6%) had done five surveys, one (0·7%) had done six surveys, and one (0·7%) had done seven surveys. 17 countries (12·1%) had done a survey at the subnational level only and ten countries (7·1%) had done more than two subnational surveys that, taken together, were extrapolated to the whole country.

Based on the most recent survey from each of the 143 countries that had done at least one survey between 2010 and 2018, 11·3% (95% CI 10·3–12·3) of boys and 6·1% (5·6–6·6) of girls reported smoking cigarettes on at least 1 day during the past 30 days, 6·0% (5·5–6·6) and 2·6% (2·4–2·9) reported smoking cigarettes on at least 3 days, and 4·2% (3·8–4·6) and 1·6% (1·4–1·8) reported smoking cigarettes on at least 6 days (table 1). The prevalence of cigarette smoking varied significantly by sex and country (figures 1, 2, appendix 2 pp 18–23). Sex differences remained similar when cigarette smoking was based on different frequency categories (ie, ≥3 days or ≥6 days during the past 30 days). The prevalence of cigarette smoking (on ≥1 day during the past 30 days) was approximately twice as high at age 15 years compared with age 13 years in both boys and girls (table 1). The prevalence of cigarette smoking (based on different frequency categories) was highest in the Western Pacific region for boys and in the European region for girls, and lowest in the African region and the Americas region for boys and in the African region and the Eastern Mediterranean region for girls (table 1). The prevalence of cigarette smoking (based on different frequency categories) generally increased with World Bank country income categories, and the association was stronger among girls than among boys. The prevalence of cigarette smoking tended to be lower in countries that had ratified the FCTC than those that had not, especially among boys (although some of the associations were not significant in girls; table 1).

The most recent overall prevalence of using tobacco products other than cigarettes (eg, chewing tobacco, snuff, dip, cigars, cigarillos, pipe, or electronic cigarettes) on at least 1 day during the past 30 days was 11·2% (95% CI 9·9–12·6) in boys and 7·0% (6·4–7·7) in girls (table 2), and the prevalence varied significantly by sex and country (figures 1, 2, appendix 2 pp 18–23). The global prevalence of using tobacco products other than cigarettes was higher at age 15 years than at age 13 years. The prevalence was highest in the Eastern Mediterranean region and lowest in the Americas

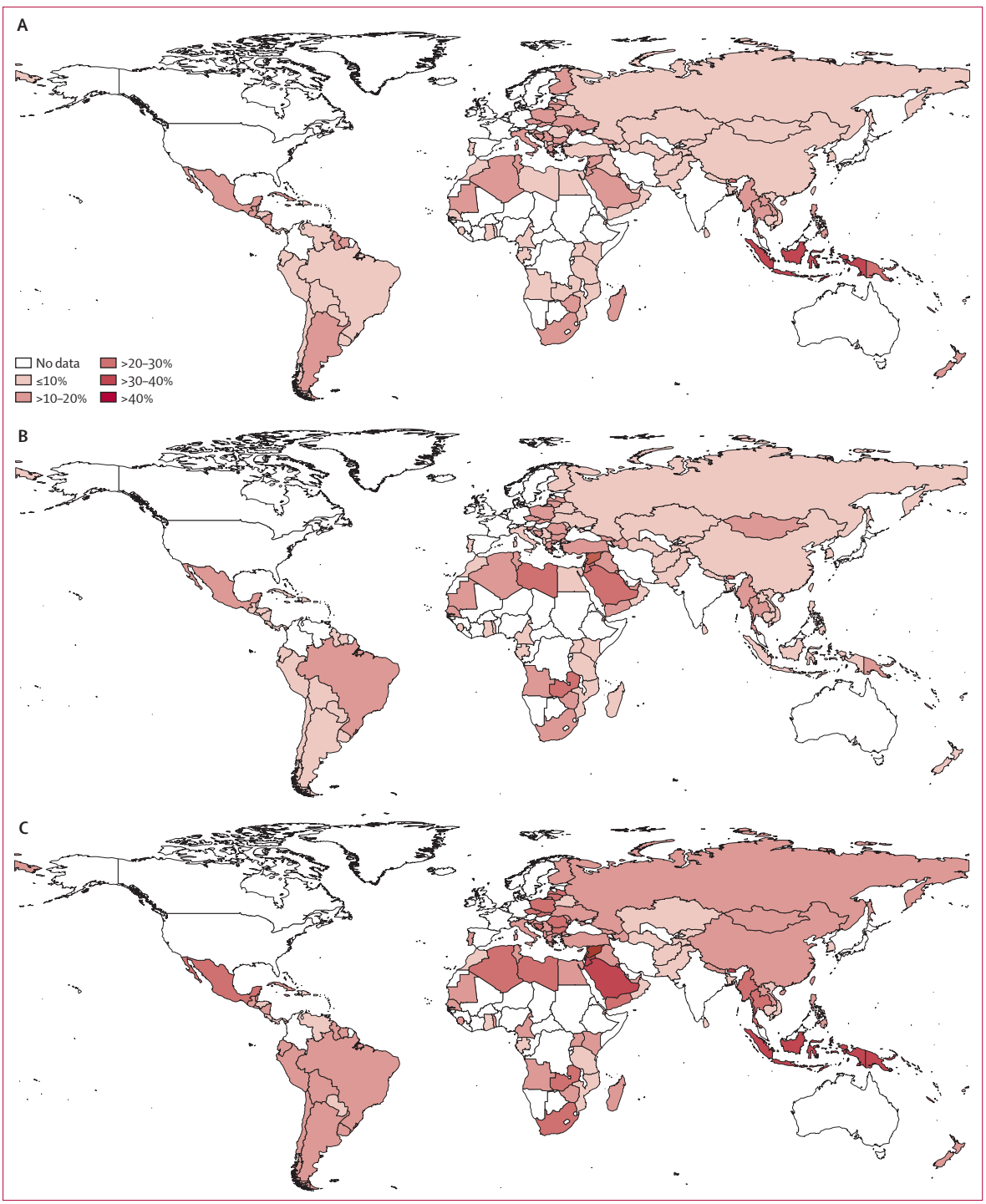


Figure 1: **Prevalence of tobacco use among boys based on the most recent Global Youth Tobacco Surveys from 143 countries, 2010–18**
(A) Cigarette smoking (ie, smoking on ≥1 day during the past 30 days). (B) Other tobacco use (ie, using a tobacco product other than cigarettes on ≥1 day during the past 30 days). (C) Any tobacco use (ie, smoking cigarettes or using other tobacco products on ≥1 day during the past 30 days).

region and in the European region (table 2). The prevalence did not differ substantially according to World Bank country income categories, or FCTC ratification status, but the prevalence tended to be lower in upper-middle-income countries (than in middle-income countries), and in countries that had ratified FCTC (than those that had not not), among both sexes (table 2).

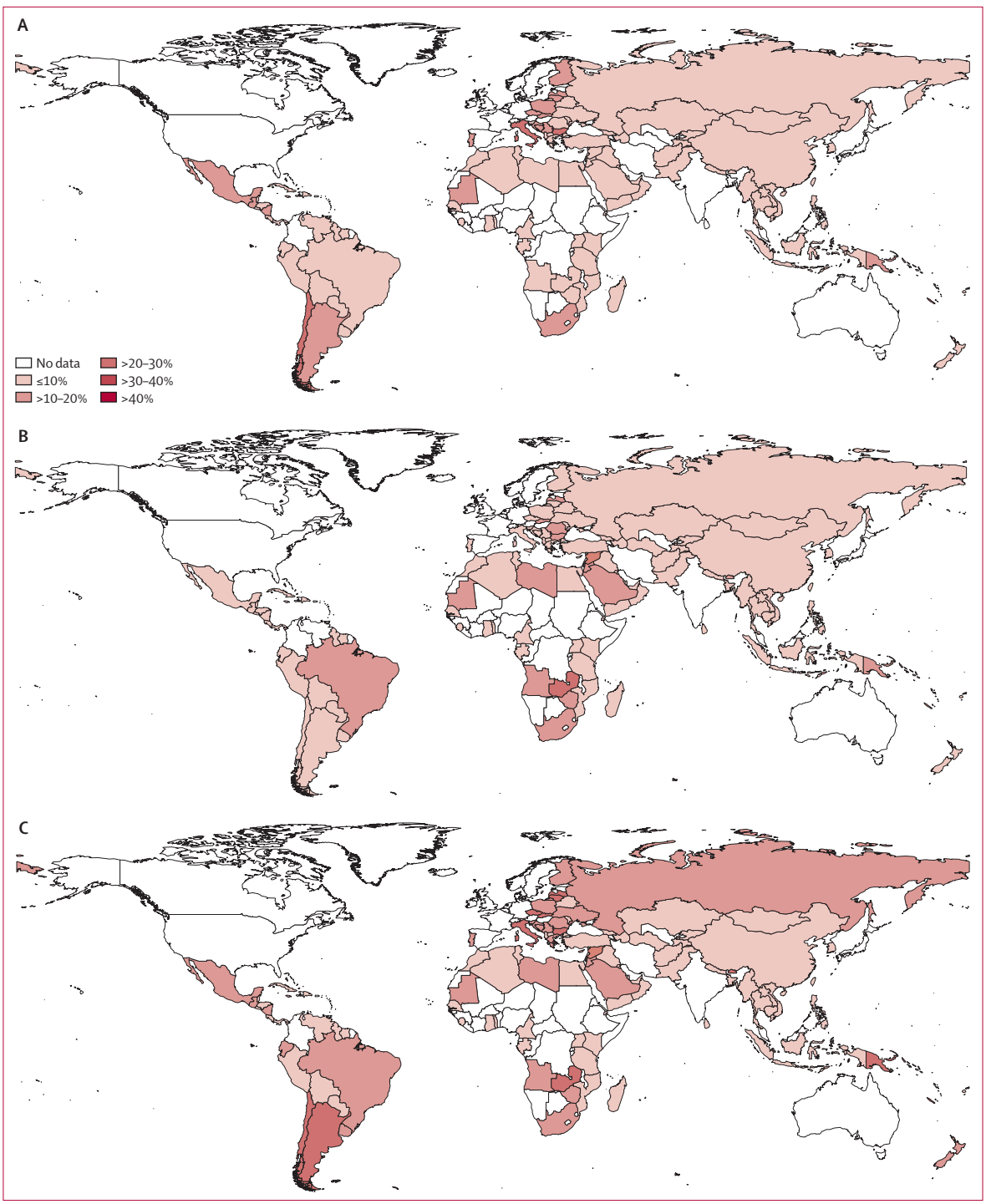


Figure 2: **Prevalence of tobacco use among girls based on the most recent Global Youth Tobacco Surveys from 143 countries, 2010–18**
(A) Cigarette smoking (ie, smoking on ≥1 day during the past 30 days). (B) Other tobacco use (ie, using a tobacco product other than cigarettes on ≥1 day during the past 30 days). (C) Any tobacco use (ie, smoking cigarettes or using another tobacco products on ≥1 day during the past 30 days).

The most recent overall prevalence of any tobacco use (ie, cigarettes or other tobacco products) on at least 1 day during the past 30 days was 17·9% (95% CI 16·1–19·6) in boys and 11·5% (10·5–12·4) in girls (table 2), and the prevalence varied significantly by sex and country (figures 1, 2, appendix 2 pp 18–23). The prevalence of cigarette smoking, use of other tobacco products, and poly use among tobacco users

	Tobacco products other than cigarettes						Any tobacco					
	Number of countries	Boys		Girls		p value (boys *vs* girls)	Number of countries	Boys		Girls		p value (boys *vs* girls)
		Prevalence	p value	Prevalence	p value			Prevalence	p value	Prevalence	p value	
Total	142	11·2% (9·9–12·6)	..	7·0% (6·4–7·7)	..	<0·0001	143	17·9% (16·1–19·6)	..	11·5% (10·5–12·4)	..	<0·0001
Age	..	..	<0·0001*	..	<0·0001*	..	..	..	<0·0001*	..	<0·0001*	..
13 years	142	8·4% (7·6–9·3)	..	5·1% (4·6–5·6)	..	<0·0001	143	12·9% (11·8–14·0)	..	8·0% (7·4–8·7)	..	<0·0001
14 years	142	10·6% (9·5–11·7)	..	6·7% (6·1–7·3)	..	<0·0001	143	17·1% (15·3–18·9)	..	11·1% (10·2–12·0)	..	<0·0001
15 years	142	13·9% (12·4–15·3)	..	9·3% (7·2–11·4)	..	<0·0001	143	23·0% (20·7–25·4)	..	15·1% (12·8–17·3)	..	<0·0001
WHO region												
Africa	22	10·7% (8·0–13·4)	0·013†	7·9% (5·5–10·3)	0·024†	0·064	22	16·2% (12·9–19·5)	0·029†	10·3% (7·3–13·2)	0·38†	0·0045
Americas	29	7·5% (6·6–8·3)	Ref	5·4% (4·7–6·0)	Ref	<0·0001	30	12·7% (11·2–14·2)	Ref	9·8% (8·6–11·0)	Ref	0·0015
Eastern Mediterranean	24	16·7% (13·4–20·0)	<0·0001†	9·0% (7·3–10·7)	<0·0001†	<0·0001	24	22·3% (18·1–26·5)	<0·0001†	11·0% (9·0–13·0)	0·16†	<0·0001
Europe	33	9·3% (5·7–12·9)	0·17†	5·5% (4·5–6·5)	0·43†	0·023	33	15·4% (11·8–18·9)	0·085†	12·0% (9·8–14·3)	0·045†	0·056
South-East Asia	9	11·8% (7·9–15·7)	0·017†	7·5% (5·1–9·8)	0·046†	0·032	9	20·9% (14·4–27·4)	0·0080†	9·7% (6·4–13·1)	0·48†	0·0013
Western Pacific	25	13·0% (10·5–15·5)	<0·0001†	8·8% (7·2–10·4)	<0·0001†	0·0030	25	23·9% (19·1–28·6)	<0·0001†	15·0% (12·5–17·5)	0·0001†	0·0006
World Bank income category												
Low	18	10·6% (7·9–13·3)	Ref	6·3% (4·9–7·7)	Ref	0·0030	18	16·6% (12·6–20·6)	Ref	8·9% (7·0–10·9)	Ref	0·0004
Middle	41	13·3% (11·0–15·5)	0·066‡	8·7% (7·2–10·2)	0·011‡	<0·0001	41	20·4% (17·0–23·8)	0·078‡	11·8% (9·7–13·9)	0·024‡	<0·0001
Upper middle	48	9·6% (8·0–11·1)	0·26‡	6·0% (5·1–6·8)	0·36‡	<0·0001	49	15·9% (13·1–18·6)	0·39‡	10·2% (8·8–11·6)	0·14‡	<0·0001
High	35	11·3% (8·1–14·5)	0·37‡	7·2% (6·0–8·4)	0·17‡	0·0090	35	18·2% (15·7–20·6)	0·25‡	14·0% (11·8–16·1)	0·0003‡	0·0058
FCTC ratification status	..	..	0·059	..	0·13	..	..	..	0·028	..	0·035	..
Not ratified	18	14·0% (10·4–17·6)	..	8·0% (6·2–9·8)	..	0·0020	18	22·6% (17·3–27·8)	..	14·0% (11·1–16·8)	..	0·0024
Ratified	124	10·9% (9·4–12·3)	..	6·9% (6·3–7·6)	..	<0·0001	125	17·2% (15·4–19·0)	..	11·2% (10·1–12·2)	..	<0·0001

Prevalence is shown as % (95% CI). Any tobacco included smoking cigarettes or using tobacco products other than cigarettes. FCTC=Framework Convention on Tobacco Control. *p value for trend across different ages. †p value compared with WHO region of the Americas as the reference. ‡p value compared with low-income countries as the reference.

Table 2: **Prevalence of use of tobacco products other than cigarettes and any tobacco use (on ≥1 day during the past 30 days) among adolescents aged 13–15 years**

also varied significantly by sex and country (appendix 2 pp 24–29).

There were marked differences in secular trends in the prevalence of cigarette smoking, other tobacco use, or any tobacco use between countries, including those within the same WHO regions (appendix 2 pp 5–17). Between 1999 and 2018 or the earliest and latest surveys, the prevalence of cigarette smoking decreased in 80 countries (57·1%), was unchanged in 39 countries (27·9%), and increased in 21 countries (15·0%; appendix 2 pp 5–11). The prevalence of cigarette smoking decreased over time in several regions, but not in the South-East Asia and Eastern Mediterranean regions; it decreased more in middle-income, upper-middle-income, and high-income countries than in low-income countries, and decreased more in countries that had ratified the FCTC than in those that had not (appendix 2 p 30). For other tobacco products, the prevalence increased in several regions, but not in the Americas and Western Pacific regions, and increased more in upper-middle-income and high-income countries than in low-income and middle-income countries (appendix 2 p 30).

The global prevalence of cigarette smoking decreased markedly over time (table 3). The linear secular trends were similar according to sex and age categories. However, the secular trends differed according to regions: the

	Number of countries	Total			Boys			Girls		
		≥1 day	≥3 days	≥6 days	≥1 day	≥3 days	≥6 days	≥1 day	≥3 days	≥6 days
Total	140	−3·5% (−4·3 to −2·7)	−2·0% (−2·7 to −1·4)	−1·3% (−1·9 to −0·8)	−4·0% (−5·0 to −3·0)	−2·5% (−3·3 to −1·6)	−1·7% (−2·4 to −0·9)	−3·4% (−4·2 to −2·5)	−2·1% (−2·8 to −1·4)	−1·0% (−1·5 to −0·5)
Age										
13 years	140	−3·7% (−5·1 to −2·3)	−2·0% (−3·1 to −0·9)	−1·2% (−2·1 to −0·2)	−4·6% (−6·3 to −2·9)	−2·4% (−3·9 to −1·0)	−1·7% (−3·0 to −0·4)	−3·2% (−4·7 to −1·8)	−2·0% (−3·2 to −0·8)	−0·9% (−1·8 to −0·1)
14 years	140	−3·4% (−4·8 to −2·0)	−2·2% (−3·3 to −1·1)	−1·5% (−2·4 to −0·6)	−3·6% (−5·3 to −1·9)	−2·2% (−3·7 to −0·8)	−1·6% (−2·9 to −0·3)	−3·3% (−4·8 to −1·9)	−2·4% (−3·6 to −1·2)	−1·7% (−2·5 to −0·8)
15 years	140	−3·3% (−4·7 to −1·9)	−1·9% (−3·1 to −0·8)	−1·3% (−2·2 to −0·4)	−3·8% (−5·5 to −2·1)	−2·7% (−4·2 to −1·3)	−1·8% (−3·0 to −0·5)	−3·5% (−4·9 to −2·1)	−1·9% (−3·1 to −0·7)	−1·6% (−2·4 to −0·8)
WHO region										
Africa	22	−3·7% (−6·3 to −1·1)	−1·7% (−3·3 to −0·1)	−0·9% (−2·2 to 0·5)	−5·4% (−8·5 to −2·3)	−2·6% (−4·8 to −0·4)	−1·6% (−3·5 to 0·2)	−2·4% (−5·7 to 0·8)	−1·2% (−2·7 to 0·3)	−0·2% (−1·4 to 0·9)
Americas	31	−4·3% (−6·5 to −2·1)	−1·9% (−3·3 to −0·5)	−1·3% (−2·5 to −0·2)	−4·5% (−7·1 to −1·8)	−2·0% (−3·8 to −0·1)	−1·3% (−2·8 to 0·3)	−5·5% (−8·4 to −2·7)	−1·9% (−3·2 to −0·7)	−1·1% (−2·1 to −0·2)
Eastern Mediterranean	25	0·1% (−2·4 to 2·4)	0·3% (−1·3 to 1·8)	0·3% (−0·9 to 1·6)	0·2% (−2·8 to 3·1)	0·4% (−1·7 to 2·4)	0·3% (−1·4 to 2·1)	−0·7% (−3·8 to 2·4)	−0·2% (−1·6 to 1·2)	0·1% (−1·0 to 1·1)
Europe	32	−5·0% (−7·2 to −2·9)	−3·6% (−5·0 to −2·2)	−2·9% (−4·0 to −1·8)	−6·2% (−8·8 to −3·6)	−4·4% (−6·2 to −2·6)	−3·4% (−4·9 to −1·9)	−4·1% (−6·8 to −1·4)	−3·0% (−4·2 to −1·8)	−2·5% (−3·4 to −1·6)
South-East Asia	10	0·5% (−3·4 to 4·3)	0·4% (−2·0 to 2·9)	0·8% (−1·2 to 2·7)	2·1% (−2·5 to 6·8)	1·3% (−2·0 to 4·5)	2·1% (−0·6 to 4·8)	−1·2% (−6·0 to 3·7)	−0·6% (−2·8 to 1·7)	−0·5% (−2·2 to 1·3)
Western Pacific	20	−6·2% (−8·9 to −3·5)	−4·2% (−5·9 to −2·4)	−2·7% (−4·1 to −1·4)	−7·7% (−11·0 to −4·5)	−6·1% (−8·4 to −3·9)	−4·3% (−6·3 to −2·4)	−5·2% (−8·6 to −1·7)	−2·8% (−4·3 to −1·2)	−1·7% (−2·9 to −0·6)
World Bank income category										
Low	20	−2·7% (−5·6 to 0·1)	−1·0% (−2·8 to 0·8)	−0·5% (−1·9 to 1·0)	−3·1% (−6·6 to 0·4)	−1·2% (−3·6 to 1·3)	−0·6% (−2·7 to 1·5)	−2·8% (−6·3 to 0·6)	−1·4% (−3·1 to 0·2)	−0·6% (−1·8 to 0·7)
Middle	39	−2·8% (−4·8 to −0·8)	−1·5% (−2·7 to −0·2)	−0·7% (−1·8 to 0·3)	−3·7% (−6·2 to −1·3)	−2·3% (−4·1 to −0·6)	−1·1% (−2·5 to 0·4)	−2·2% (−4·7 to 0·3)	−0·8% (−1·9 to 0·3)	−0·5% (−1·3 to 0·3)
Upper middle	50	−3·4% (−5·2 to −1·6)	−1·7% (−2·9 to −0·6)	−1·2% (−2·1 to −0·3)	−3·5% (−5·7 to −1·3)	−2·0% (−3·5 to −0·4)	−1·6% (−2·8 to −0·3)	−3·4% (−5·6 to −1·2)	−1·7% (−2·6 to −0·7)	−1·0% (−1·7 to −0·3)
High	31	−5·2% (−7·5 to −3·0)	−3·9% (−5·3 to −2·4)	−3·0% (−4·1 to −1·8)	−6·5% (−9·3 to −3·7)	−4·7% (−6·6 to −2·7)	−3·5% (−5·2 to −1·9)	−5·6% (−8·4 to −2·9)	−3·5% (−4·7 to −2·2)	−2·6% (−3·6 to −1·6)
FCTC ratification status										
Not ratified	20	−2·4% (−5·2 to 0·5)	−1·4% (−3·3 to 0·4)	−1·0% (−2·4 to 0·5)	−2·3% (−5·8 to 1·2)	−1·8% (−4·2 to 0·7)	−1·3% (−3·3 to 0·8)	−3·2% (−6·6 to 0·3)	−1·6% (−3·2 to −0·1)	−1·0% (−2·2 to 0·2)
Ratified	120	−3·7% (−4·9 to −2·6)	−2·1% (−2·9 to −1·4)	−1·4% (−2·0 to −0·8)	−4·5% (−5·9 to −3·1)	−2·7% (−3·7 to −1·7)	−1·8% (−2·6 to −0·9)	−3·5% (−4·9 to −2·1)	−1·8% (−2·5 to −1·2)	−1·2% (−1·7 to −0·7)

Data are n or % (95% CI). FCTC=Framework Convention on Tobacco Control.

Table 3: **Linear trends per 10 calendar years in the prevalence of cigarette smoking among adolescents aged 13–15 years, between 1999 and 2018 or the earliest and latest surveys, by sex and frequency of cigarette smoking (number of smoking days during the past 30 days)**

prevalence of cigarette smoking decreased more in the Western Pacific, European, Americas, and African regions (a decrease of more than 3·5% per 10 calendar years based on cigarette use on ≥1 day during the past 30 days). However, the prevalence of using tobacco products other than cigarettes tended to increase globally between the earliest and latest surveys in each country (0·6% increase; 95% CI −0·8 to 1·9; appendix 2 p 31). The increase was significant in boys aged 15 years. The prevalence of using tobacco products other than cigarettes increased in boys in the Eastern Mediterranean region, but decreased among both sexes in the Western Pacific region. Trends in the prevalence of using tobacco products other than cigarettes were not associated with World Bank income categories or FCTC ratification status (appendix 2 p 31).

The secular trends for cigarette smoking and the trends for using other tobacco products tended to have opposite directions in several countries. Therefore, some of the linear regression estimates for the secular trends in the prevalence for any tobacco use (ie, cigarette smoking or using other tobacco products; appendix 2 p 32) are of a lesser magnitude than either the secular trends for cigarette smoking or for the use of other tobacco products.

Discussion

Based on data from 143 countries that had done at least one survey between 2010 and 2018, the most recent global prevalence of cigarette smoking was approximately 11·3% in boys and 6·1% in girls aged 13–15 years. The most recent prevalence was higher in high-income

countries than in low-income countries. The most recent prevalence of using tobacco products other than cigarettes was as high as the prevalence of cigarette smoking. We found that the prevalence of cigarette smoking decreased in the majority of countries in this study, and the decrease tended to be larger in countries that had ratified FCTC than in those that had not. However, the prevalence of using tobacco products other than cigarettes increased in the majority of countries.

This is the first study, to our knowledge, to assess the prevalence of cigarette smoking according to different smoking frequencies. The prevalence of cigarette smoking was substantially lower when estimates were based on smoking cigarettes on at least 3 days or at least 6 days versus at least 1 day during the past 30 days. The prevalence based on smoking on several days per month could better reflect already sustained smoking habits and addiction as opposed to mere experimentation. However, as many as 10% of adolescents who started to smoke cigarettes at irregular intervals became regular smokers after 1 year of follow-up.[10] Health education programmes and regulatory tobacco control measures to prevent the uptake of tobacco use among adolescents remain crucially important to prevent tobacco use at an early age and subsequent further tracking into adulthood in view of the addictive nature of tobacco products.

The prevalence of cigarette smoking and of the use of other tobacco products were two or three times higher at age 15 years than at age 13 years in most countries, consistent with rapidly increasing tobacco uptake as adolescents get older, which has been shown in many previous studies.[8,11,12] The increasing prevalence of tobacco use with increasing age among adolescents might partly be explained by increasing risk-taking in adolescence and a desire for new experiences during transition from childhood to adulthood.[13] Peer pressure and adolescents' economic ability to buy cigarettes can also be important factors in tobacco use among young people.[14] Tobacco advertising, which portrays smoking as a sign of freedom, success, and adulthood, has an important role among adolescents.[15] This advertising includes tobacco promotion and sponsorship (eg, sponsored concerts and cultural events) that are appealing to young people where adolescents have opportunities to experiment with cigarette smoking.[16] Fortunately, many countries are increasingly implementing partial or total bans on tobacco advertising, promotion, and sponsorship, as required by the FCTC. It is, however, challenging to address advertisement and promotion of tobacco products in new so-called under-the-radar media, such as the internet and social networks.[17,18] More generally, the sharp increase in the prevalence of tobacco use according to age stresses the need for health education starting at an early age and adequate regulatory tobacco control interventions.

Different ratios between sexes in the prevalence of tobacco use across countries might also reflect different cultural and social tolerance to tobacco use. For example, an extreme difference between sexes was found in China, where as many as 55% of men but only 4% of women were reported to smoke cigarettes.[19] As a low prevalence of tobacco use among women (including female adolescents) constitutes a large potential market for the tobacco industry,[20] efforts to reduce the uptake of tobacco products are particularly important in countries where tobacco use has traditionally been low.

The prevalence of cigarette smoking decreased in the majority of the countries included in this study, for example in Burkina Faso, Namibia, Uruguay, Somalia, Belarus, Cook Islands, and Papua New Guinea. These findings are consistent with results from a previous review of GYTS in 1999–2008, which showed that among 100 countries surveyed, 27 countries (27%) reported a decrease in tobacco use among adolescents, 61 countries (61%) reported no change, and ten countries (10%) reported an increase.[21] Our study extends trend estimates reported in these previous reviews to more recent years (2010–18), showing that among the 140 countries included in our analysis of trends, the prevalence of cigarette smoking decreased in 80 countries (57·1%), was unchanged in 39 countries (27·9%), and increased in 21 countries (15·0%). Differences in upward versus downward trends in the prevalence of tobacco use between different countries or regions might reflect differences in the implementation of the key tobacco control measures advocated by the FCTC. Other factors that can account for differences in the trends in tobacco use between countries include the commitment of governments and engagement of civil society and non-governmental organisations, as described, for example, in Papua New Guinea,[22] or in several Pacific islands.[23] Similarly, Uruguay has been at the forefront of tobacco control, with a complete ban on tobacco promotion and advertisement, stringent requirements on pictorial health warnings, and provision of treatment of nicotine dependence at the health-care level.[24] However, although the prevalence of cigarette smoking decreased globally between 1999 and 2018, a challenge is that the prevalence of using other tobacco products increased, suggesting a transition from cigarette smoking to using other tobacco products (eg, electronic cigarettes).

Worryingly, the prevalence of cigarette smoking did not decrease in several countries in the African, European, Eastern Mediterranean, and South-East Asia regions (eg, Botswana, Zimbabwe, Bosnia and Herzegovina, and Indonesia). Possible factors underlying the increasing prevalence of smoking in these countries might include a low cost of cigarettes due to low excise tax on tobacco products; loose or poorly enforced smoking bans in enclosed places; weak measures to ban tobacco advertisement, promotion, and sponsoring; continued sales of cigarettes by single sticks; local production of tobacco and tobacco products, which can undermine national tobacco control responses; absence of ratification or

weak implementation of the FCTC; and continued social tolerance of tobacco use.[25]

We found that the prevalence of smoking among adolescents tended to decrease more in countries that had ratified the FCTC than in countries that had not. However, the association was not large, possibly because the sole information on FCTC ratification status does not equate to actual implementation and enforcement of the regulatory measures prescribed by the FCTC at the national level or, similarly, advised by the WHO MPOWER package. However, our finding is consistent with a lower demand for tobacco products, including among adolescents, in countries where provisions of the FCTC have been implemented.[21,26] The FCTC and the related MPOWER package have dramatically changed tobacco control efforts worldwide in the past two decades.[27]

For more on **MPOWER** see https://www.who.int/tobacco/mpower/en/

The prevalence of using tobacco products other than cigarettes was as high as, or higher than, the prevalence of cigarette smoking in many countries. Waterpipe use has re-emerged globally[28] and can be appealing to adolescents,[29,30] especially in the Eastern Mediterranean region and in eastern Europe.[31] However, the question on tobacco use in the GYTS questionnaire does not enable the identification of which tobacco products other than cigarettes are used in a particular country. Future surveys should include questions on specific other tobacco products (including electronic cigarettes). It is crucial to assess which other tobacco products adolescents predominantly use in order to plan tobacco control efforts accordingly, as different policies might be needed for different tobacco products.

For more on **US Centers for Disease Control and Prevention** see https://www.cdc.gov/tobacco/global/gtss/gtssdata/index.html

Our study has several strengths. We used the most recent available data from a large global tobacco use surveillance instrument in a large number of countries. The same sampling strategy and standard questionnaire were used in all countries, making results directly comparable between countries and over time. We also assessed the prevalence of different frequencies of cigarette smoking (ie, on ≥3 days or ≥6 days during the past 30 days), which might better represent the prevalence of regular cigarette smoking. However, this study also has several limitations. First, data were self-reported, with possible over-reporting or under-reporting. However, the anonymous nature of the questionnaire might have minimised these concerns, and good test–retest reliability has been reported with the questions about tobacco use in the GYTS.[32] Second, some low-income and middle-income countries were not included in our review because they had not done a GYTS in the time intervals considered in this study. Third, the question "During the past 30 days, on how many days did you smoke cigarettes?" only asked about frequency of smoking days during the past 30 days, and did not allow assessment of whether an adolescent would smoke a cigarette or another tobacco product fully or, for example, share it with others. Fourth, among the 140 countries included in the trend analysis, 53 countries had done only two surveys (as opposed to more than two surveys), which weakens our ability to assess robust linear trends. Additionally, the study years for the first survey and for the last survey across all included countries were largely different. Fifth, there is still no common metric for publicly assessing the FCTC implementation, which needs more attention in future studies.

Tobacco use among young adolescents remains an important public health challenge worldwide. Although the prevalence of cigarette smoking decreased over time in the majority of countries in this study, the prevalence of the use of tobacco products other than cigarettes increased in the majority of countries. These results re-emphasise the importance of regular monitoring of the prevalence of tobacco product use in populations, including among adolescents, and the need to strengthen tobacco control efforts globally. These findings echo WHO's 2020 World No Tobacco Day campaign, which focuses on protecting children and young people from exploitation by the tobacco and related industries.[33]

Contributors

BX and PB designed the study and led the writing of the paper. BX was the principal investigator. CM did the data analysis and drafted the first version of the manuscript. BX and CM accessed and verified the data. All authors critically revised the manuscript and approved the final version of the manuscript.

Declaration of interests

We declare no competing interests.

Data sharing

Data collected for this study, including de-identified individual participant data and a data dictionary defining each field in the dataset, as well as additional related documents such as study protocol, will be available to others upon publication, without any restrictions, from the US Centers for Disease Control and Prevention website.

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